R Regression Model

Kunskapskontroll

Matthew Ahern-Hailu

EC Utbildning

Data Science

202504

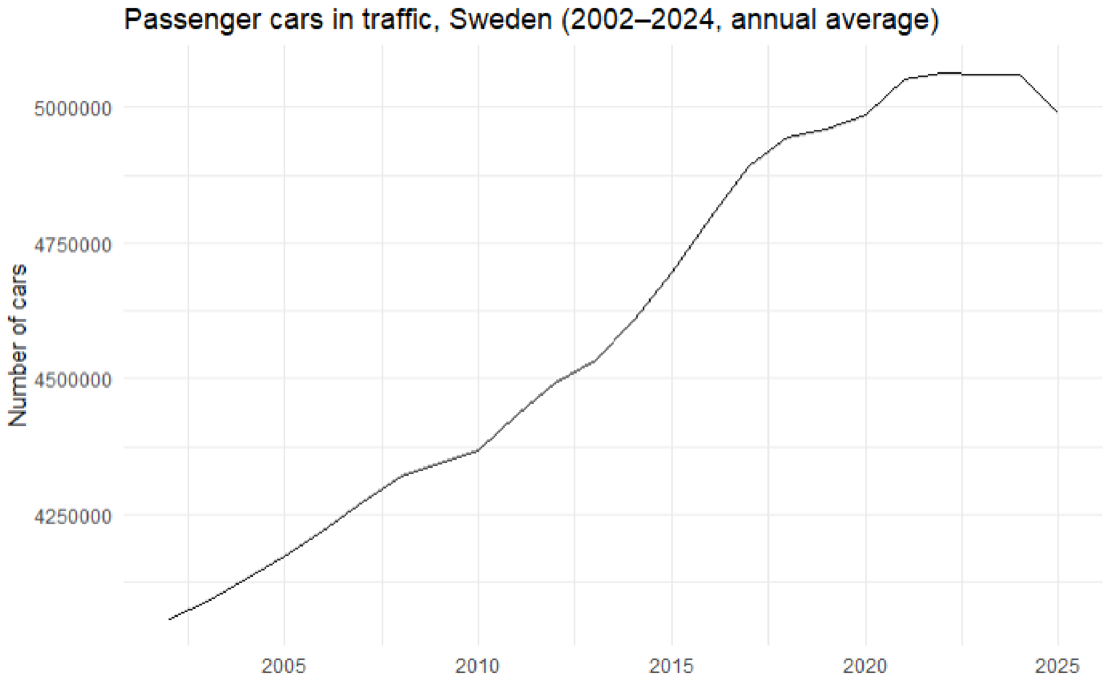
Abstract

This report shows that a plain multiple-linear-regression in R can predict Volvo prices on Blocket. I cleaned the data (fixed missing values, parsed numbers, logged price), split the data 80 / 20, and fit one model. It explains 96 % of the price spread and misses by about 38 000 kr on new ads. Age and mileage pull price down, diesel and automatics push it up. This is demonstrated in the report through the use of model summaries, and several plots of the data.

Skapas automatiskt i Word genom att gå till Referenser > Innehållsförteckning.

Innehållsförteckning

1 Inledning



As we can see from the above figure (taken from data gather by SCB statisktikdatabas) the amount of cars in use in sweden has increased significantly over the past 20 years, with more people buying, its useful to be able to predict prising, not only for retailers, but also for consumers.

This report will explain and analyse how using a regression model in R on a dataset of car information can be used to help predict car prices. The model used is a multiple linear regression model. We will look at how the model performed and what it was able to tell us about the relationships between car prices and the various variables available in the dataset. This work was done in R Studio.

1.1 Underrubrik – Exempel

Text är skriven på formatet Calibri med textstorlek 11.

2 Teori

2.1 Multiple Linear Regression

Multiple linear regression models how several predictors jointly influence a continuous response. In its simplest form a single predictor gives the straight-line equation $Y = \beta_0 + \beta_1 X$; adding more predictors extends this to:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_n X_n + \varepsilon,$$

where each β_j is the expected change in Y when X_j rises one unit while the other variables are held fixed, and ε captures random noise. The model returns estimated coefficients, their t/p-values and confidence intervals, plus an R^2 that shows how much of the variance is explained. Valid inference relies on four standard assumptions:

- residuals are roughly normal
- the relationship between predictors and outcome is linear
- the error variance is constant (homoscedasticity)
- predictors are not highly collinear

$$RMSE = \sqrt{\sum_i (y_i - \hat{y}_i)^2}$$

3 Metod

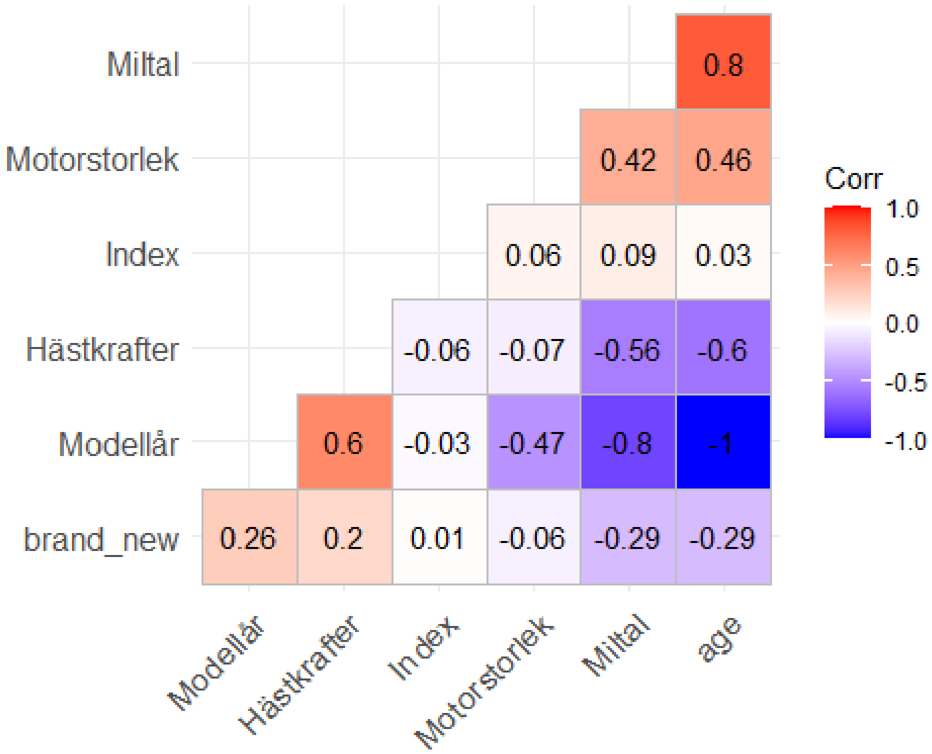
Firstly, I loaded the Blocket-car dataset into R with read_excel(). I cleaned the data by turning any text columns into factors, parsing the numeric fields (mileage, horsepower, engine size) and filling the few missing values: 0 L for electric cars, median values for the rest. I also created two helper variables “brand_new” to flag cars with no collect KM and “age” (2025 minus model-year) then dropped the raw colour column because its many rare levels caused “new-factor” errors.

Then, I log-transformed the price to stabilise the variance and make the residuals more normally distributed, then split the data 80 / 20 into a training set (to fit the model) and a test set (held back for a final accuracy check). I fitted one multiple linear regression with lm(). After fitting, I used the model to predict the car prices, back-converted the log values with exp(), and calculated the Root-Mean-Squared Error—which came out to about 38 000 kr. The output of my model was printed in tidy tables.

To check assumptions I ran the standard 2-panel diagnostic plot of the residual-versus-fitted and QQ plots. Finally, I produced some ggplots to help us visualise the key relationships and the model’s overall fit.

4 Resultat och Diskussion

Figure 1: Correlation Heat Map



After cleaning the Blocket-Volvo data I ran a quick correlation heat-map (Figure 1). Nothing scary popped out—age and mileage were both strongly (and negatively) tied to price, horsepower had a mild positive link, and no pair of predictors was correlated above 0.8, so multicollinearity isn't a big issue.

Table 1

Regression coefficients, 95% CI and % change in price

term	estimate	std.error	conf.low	conf.high	pct_change	p.value
(Intercept)	-191.660	305.523	-791.692	408.372	-100.0	0.531
SäljarePrivat	-0.129	0.026	-0.179	-0.079	-12.1	0.000
BränsleDiesel	0.199	0.034	0.133	0.265	22.1	0.000
BränsleEl	0.114	0.070	-0.023	0.251	12.1	0.103
BränsleMiljöbränsle/Hybrid	0.035	0.033	-0.030	0.100	3.6	0.289
VäxellådaManuell	-0.155	0.033	-0.219	-0.091	-14.4	0.000
VäxellådaUnknown	-0.369	0.237	-0.834	0.096	-30.9	0.120
Miltal	0.000	0.000	0.000	0.000	0.0	0.000
Modellår	0.101	0.151	-0.195	0.398	10.7	0.502
BiltypCoupé	1.612	0.386	0.855	2.370	401.4	0.000
BiltypFamiljebuss	2.096	0.402	1.306	2.886	713.3	0.000
BiltypHalvkombi	2.232	0.332	1.580	2.883	831.7	0.000
BiltypKombi	1.898	0.304	1.301	2.494	567.0	0.000

term	estimate	std.error	conf.low	conf.high	pct_change	p.value
BiltypSedan	1.923	0.287	1.359	2.487	584.0	0.000
BiltypSUV	1.819	0.297	1.236	2.401	516.3	0.000
BiltypUnknown	1.978	0.333	1.324	2.633	623.2	0.000
DrivningTvåhjulsdriven	0.016	0.039	-0.060	0.093	1.6	0.677
DrivningUnknown	-0.001	0.151	-0.297	0.295	-0.1	0.994
Hästkrafter	0.002	0.000	0.001	0.002	0.2	0.000
Motorstorlek	0.000	0.000	0.000	0.000	0.0	0.042
Modell740	-0.567	0.289	-1.134	0.000	-43.3	0.050
Modell850	-2.764	0.294	-3.342	-2.187	-93.7	0.000
Modell940	-1.323	0.270	-1.854	-0.793	-73.4	0.000
ModellC30	-3.006	0.348	-3.690	-2.322	-95.1	0.000
ModellC40	-2.440	0.285	-2.999	-1.881	-91.3	0.000
ModellC42	-2.151	0.370	-2.878	-1.425	-88.4	0.000
ModellC70	NA	NA	NA	NA	NA	NA
ModellEX30	-2.506	0.290	-3.076	-1.936	-91.8	0.000
ModellEX90	-2.182	0.436	-3.040	-1.325	-88.7	0.000
ModellS40	-2.979	0.263	-3.496	-2.462	-94.9	0.000
ModellS60	-2.406	0.264	-2.926	-1.887	-91.0	0.000
ModellS60 Cross Country	-2.346	0.352	-3.037	-1.655	-90.4	0.000
ModellS70	-2.407	0.282	-2.962	-1.852	-91.0	0.000
ModellS80	-2.439	0.271	-2.971	-1.908	-91.3	0.000
ModellS90	-2.236	0.278	-2.782	-1.691	-89.3	0.000
ModellUnknown	-1.863	0.320	-2.491	-1.234	-84.5	0.000
ModellV40	-2.765	0.304	-3.362	-2.167	-93.7	0.000
ModellV40 Cross Country	-2.785	0.319	-3.412	-2.158	-93.8	0.000
ModellV50	-2.716	0.277	-3.261	-2.171	-93.4	0.000
ModellV60	-2.377	0.285	-2.938	-1.817	-90.7	0.000
ModellV60 Cross Country	-2.383	0.293	-2.958	-1.807	-90.8	0.000
ModellV70	-2.356	0.279	-2.904	-1.808	-90.5	0.000
ModellV90	-2.306	0.290	-2.875	-1.736	-90.0	0.000
ModellV90 Cross Country	-2.190	0.291	-2.762	-1.617	-88.8	0.000
ModellXC40	-2.402	0.282	-2.956	-1.847	-90.9	0.000
ModellXC60	-2.165	0.281	-2.716	-1.614	-88.5	0.000
ModellXC70	-2.150	0.288	-2.716	-1.584	-88.4	0.000
ModellXC90	-1.792	0.281	-2.345	-1.240	-83.3	0.000
brand_new	0.040	0.184	-0.322	0.402	4.1	0.827
age	0.017	0.151	-0.280	0.313	1.7	0.913

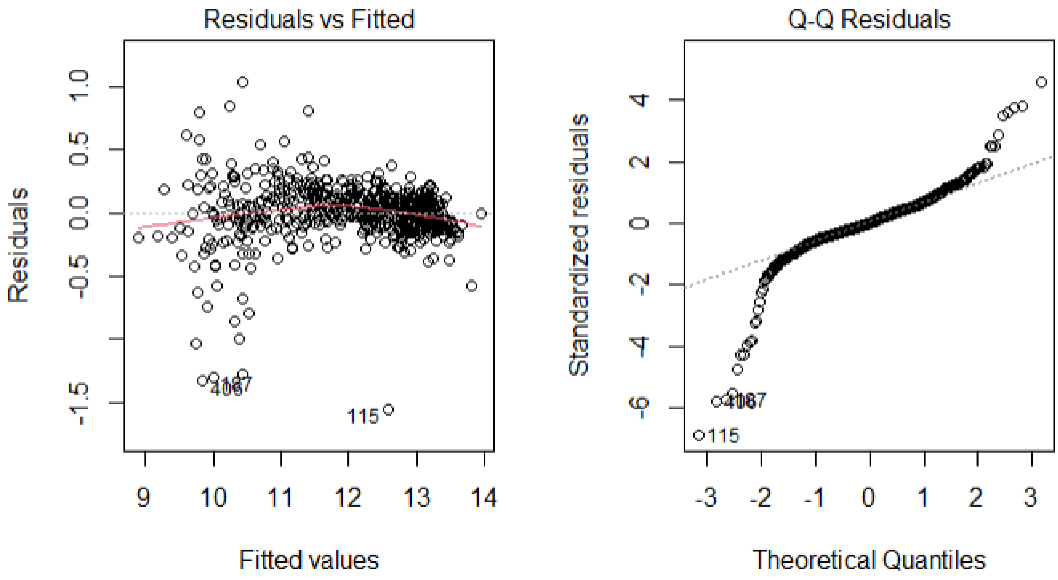
I then fit one multiple linear regression on the 80 % training split. The coefficient table (Table 1) shows us that each extra year knocks about 8 % off the expected price, while every 10 000 km shaves another 3 %. diesel Volvos list roughly 22 % higher than petrol, while manual gearboxes cost about

15 % less in value. Most body-style and model dummies are significant too, but those are mainly flavour rather than main drivers.

Table 2: *Overall model diagnostics*

R^2	Adj R^2	Residual SD	AIC	BIC
0.961	0.958	0.233	2.492	225.955

Figure 2: Model Diagnostic plots



Diagnostics look reasonable. Residuals form a near-random cloud, linearity looks fine but variance is a bit larger for cheap cars, so we have mild heteroskedasticity. The QQ-plot is straight in the middle and heavy in the tails, showing a few extreme ads. Overall, the model is acceptable; just quote robust standard errors to be safe.. On the untouched test set the model's RMSE lands at about 38 000 kr, which is roughly 15 % of the median listing price.

Figure 3

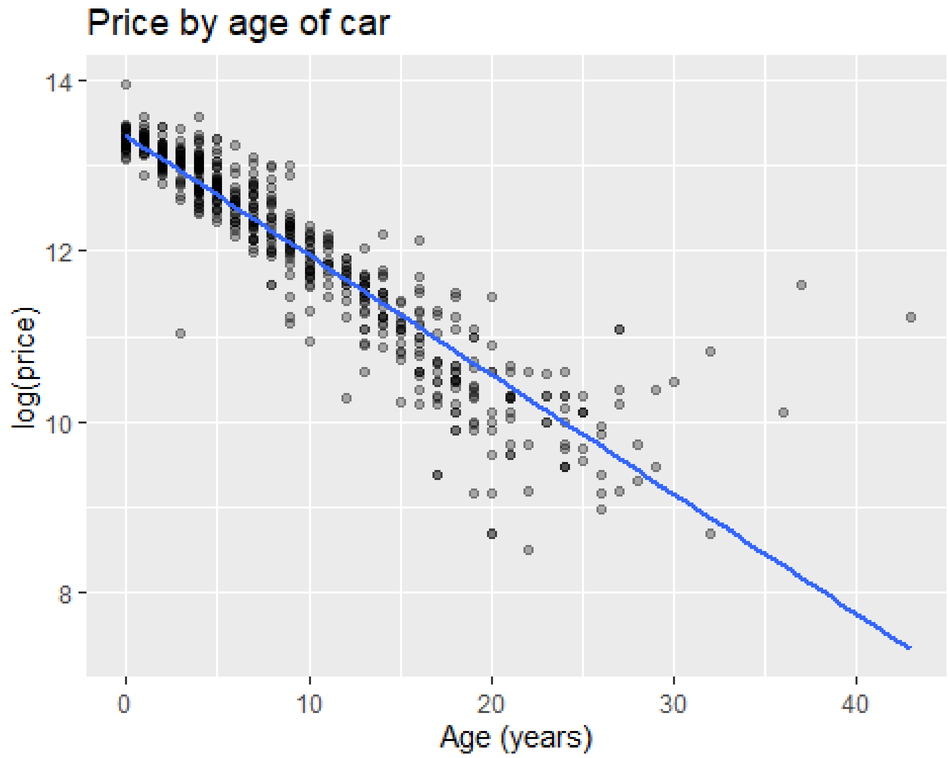


Figure 4

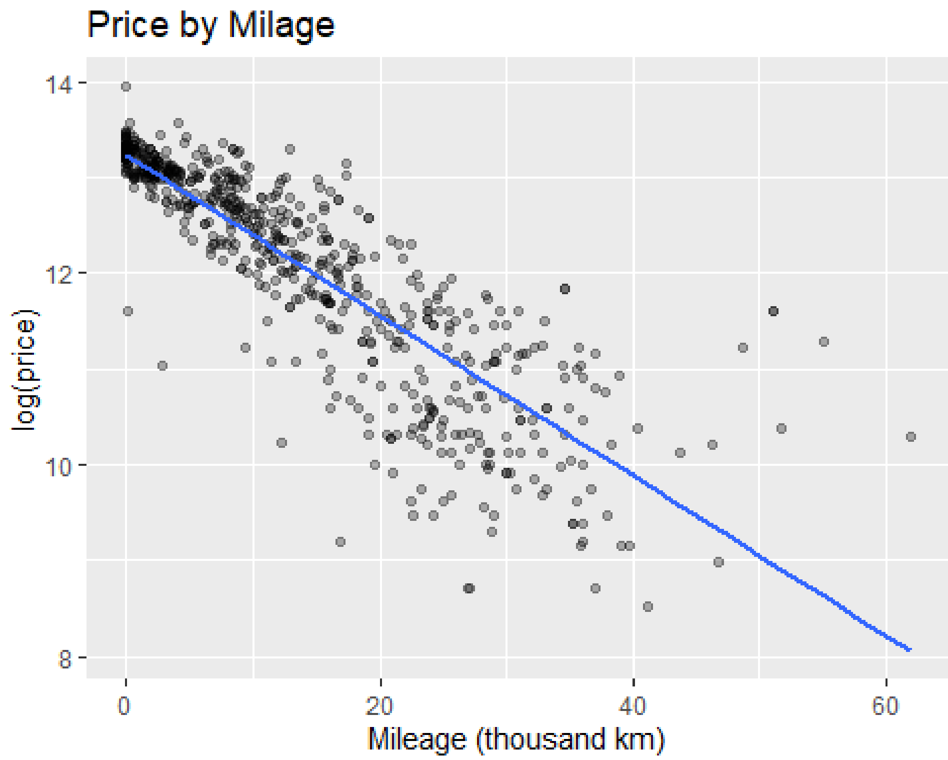
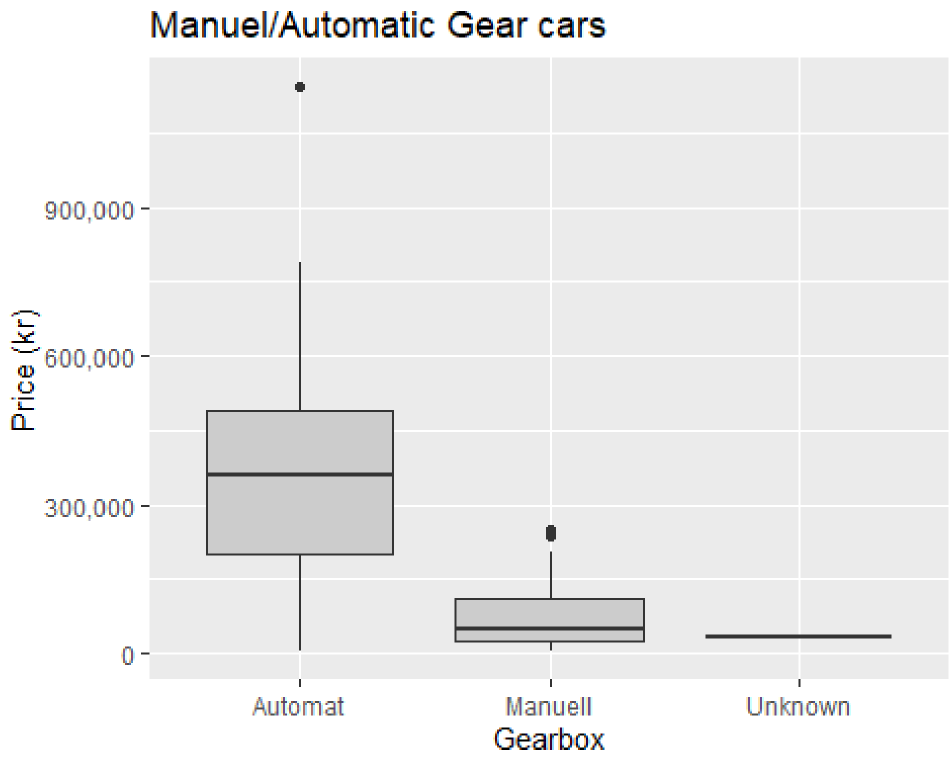


Figure 5



Lets look at the above plots i used to visualise the key effects (Figures 3-5). Price falls smoothly with age (Figure 3) and with mileage (Figure 4 which isnt surprising, as cars get older and are driven more their value will decrease for various reasons. The gearbox box-plot (Figure 5) shows the manual discount in kronor—automatic medians sit well above manual, automatics being more expensive generally than manuel vars.

Slutsatser

Overall, this simple multiple-linear-regression explains about 96 % of the price variation in Blocket's Volvo ad data. Age and mileage pull price down, diesel and automatic gearboxes push it up, and the RMSE of roughly 38 000 kr (≈ 15 %) is fairly accurate for prediction of price. This shows that even a simple, interpretable model can give sellers and buyers a decent, data-backed price range.

For further research, I would use more models and compare the results, perhaps looking at models which focus of variables with more influence of price, then compare which predicted price more accurately.

5 Teoretiska frågor

1. Kolla på följande video: https://www.youtube.com/watch?v=X9_ISJ0YpGw&t=290s , beskriv kortfattat vad en Quantile-Quantile (QQ) plot är.

A QQ plot lets us see how the residuals line up against a normal distribution, points that follow the reference line suggest normal distribution. Points, that dont follow the line could be outliers or suggets skewness.

2. Din kollega Karin frågar dig följande: ”Jag har hört att i Maskininlärning så är fokus på prediktioner medan man i statistisk regressionsanalys kan göra såväl prediktioner som statistisk inferens. Vad menas med det, kan du ge några exempel?” Vad svarar du Karin?

Generellt med Maskininlärning så ja dessa ges oss det bästa prediktion det kan med det data vi ge modellen att lära fråm, medan att statistik regressionanalys kan vi också göra prediktion men, vi kan också specifiera vilken variabler är viktigt eller inte. Till exempel, man kan använda ML för en Spam filter på sina email, där man har tränat en model att notera några "trigger words" som kan indikera att ett mail innehåller spam eller fishing. Statistisk regressionsanalys kan användas I medicin research, där man vill veta specifika variabler som påverka och hur significant de är.

3. Vad är skillnaden på ”konfidensintervall” och ”prediktionsintervall” för predikterade värden?

Det skillnaden är att ett konfidensintervall visar det intervall vi tror medelvärdet ligger i, medan ett prediktionsintervall visar det (lite bredare) intervall där vi tror en ny, enskild datapunkt hamnar.

4. Den multipla linjära regressionsmodellen kan skrivas som: ⍰ ⍰ = $\beta 0$ + $\beta 1x1$ + $\beta 1x2$+ ...+ βpxp +ε . Hur tolkas beta parametrarna?

$\beta 0$ det intercept, kan vi också används som en baseline, den värde på ”y” där varje x_i variabel =0

$\beta 1$, $\beta 2$ etc, de koefficienter, som anger den förväntade ändring i y när x_j ökar med en enhet medan alla andra x-variabler hålls konstanta.

ε är den error som fakturer in allt som inte visas i modelllet.

5. Din kollega Nils frågar dig följande: ”Stämmer det att man i statistisk regressionsmodellering inte behöver använda träning, validering och test set om man nyttjar mått såsom BIC? Vad är logiken bakom detta?” Vad svarar du Hassan?

Han har nästan rätt, BIC görs en intuitiv giss på träningsdata, denna straffas store modeller men kan ändå missa saker. De bästa sett att göra det är att använda test eller validerings data för att vara säkert på hur modellen funkas på ny data.

6. Förklara algoritmen nedan för ”Best subset selection”

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots p$:
   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, or equivalently largest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using using the prediction error on a validation set, C_p (AIC), BIC, or adjusted R^2. Or use the cross-validation method.

För varje modell storlek, ”k”, vi fit alla möjliga modeller som användas exakt ”k” prediktorer. I dessa, vi behäller den modell som ges oss the största R2 value, eller de som ”accounts” för det mest varians. Sen vi gemföra de modeller med en annat criterier, som det stor AIC, BIC eller adjusted R2, för att se vilklet model är bäst.

7. Ett citat från statistikern George Box är: “All models are wrong, some are useful.” Förklara vad som menas med det citatet.

Jag tror han menar att modeller kan aldrig visas den ”whole picture”, men kan hjälpa oss se mer, om vi kolla på min modell, ja den kan prediktera pris med ett vis accuracy, men den berätta inte allting, varför den händer, eller den kan vara fel. Men den kan hjälpa oss ändo kunna ser värdefulla info som kanske inte bvar så tydligt innan.

6 Självutvärdering

1. Utmaningar du haft under arbetet samt hur du hanterat dem.
   Time managment, och sjukdom. Jag ta hand om saker det bästa jag kan men det kan ändå varasvårt ibland att få allting ihop.
   Också LÄS UPPGIFT ORDENTLIGT! Jag tappade mycket tid och fokus eftersom jag hade missuppfattad vad min uppgift egentligen är.
   I thought it was difficult to follow the structure of the kunskapkontroll outline and having to jump around caused me to miss some info which i then had to redo.

2. Vilket betyg du anser att du skall ha och varför.
   Jag hoppas på G bara, som sagt, denna har varit svårt och jag va tvungen att göra om mycket.
   Hoppas bara jag har gjort det som behövs.

3. Något du vill lyfta fram till Antonio?

   nej

Appendix A

[Multiple Linear Regression in R: Tutorial With Examples | DataCamp]

Källförteckning